

Thread Display: A Novel Wind-Driven Spatial Display Technique

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Figure 1: (a) Actual projection onto the movable thread display. (b) System overview: laser projector (left), thread display (right), and Aeroar unit (bottom).

CCS Concepts

• Human-centered computing → Displays and imagers.

Keywords

Spatial display, Wind-driven display, Retroreflective thread, Real-time dynamic screen

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1 Abstract

This study proposes the “Thread Display”, a novel spatial display that forms a screen in mid-air from wind-suspended, retro-reflective polyethylene threads. Unlike particle-based displays, our primary innovation is a real-time control system that shapes the screen to the content’s silhouette. This retraction capability minimizes visual intrusion and enhances the floating effect. By eliminating material replenishment, our method advances the operational efficiency, safety, and flexibility of aerial displays.

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2 Introduction

Spatial displays present information in real space without physical screens. Conventional methods like Fogdisplays[Suzuki et al. 2017] use particles but suffer from low resolution, complex installation, and instability. More recent approaches like FlexiVol and ProjecString[Bouzbib et al. 2025; Chang et al. 2022] or holographic displays[Geng 2013] are often costly or require special environments.

In contrast, our proposed Thread Display avoids consumables by using wind-suspended, retro-reflective threads. Real-time height control via servo motors and an Arduino, guided by Unreal Engine’s contour analysis, retracts unused screen portions (Figure 2). This approach offers improved practicality and potential for multi-sensory interactions, such as haptic feedback.

3 Method

The core element is a **polyethylene thread coated with retro-reflective paint**. We chose polyethylene for its lightweight, durable, and low-cost properties, which ensure implementation stability. Each thread is anchored at its base and suspended by an upward airflow from a air blower, forming a screen in mid-air (Figure 3). This design eliminates the need for consumable media and reduces environmental dependencies. The figure shows the composition of the projection system(Figure 1). We use a Sony Pico MPCL1 Mobile Projector for its focus-free laser projection. Height control for the 9 thread rows is managed by **servo motors and an Arduino**. Each thread is assigned a motor for individual control, allowing the screen’s shape to be varied dynamically (Figure 4, 5). A flexible, clear partition prevents thread entanglement with the slide rail. The system connects to Unreal Engine 5.4 to dynamically acquire the maximum Z-axis height of a 3D model. This data is linked to the

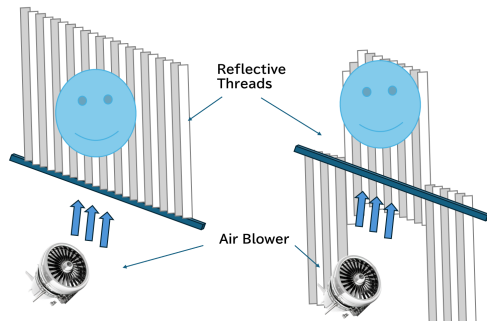


Figure 2: Contour analysis of a 3D model and retraction of unnecessary areas using Unreal Engine 5.4. The thread length is automatically adjusted to the model's highest point on the Z-axis.

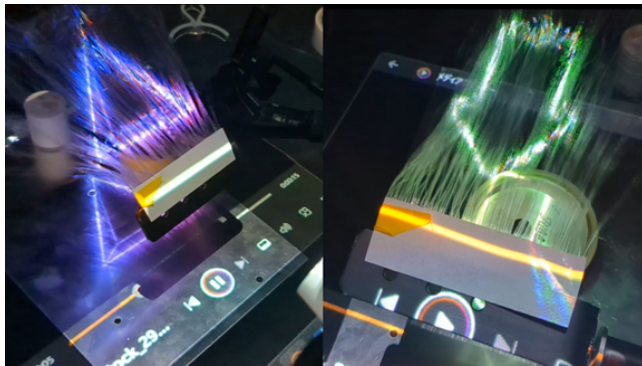


Figure 3: Projection Experiment on Retro-Reflective Threads.

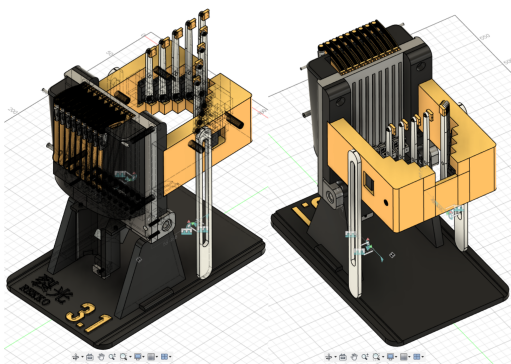


Figure 4: A proprietary device was developed to control the threads via servo motors.

thread control parameters, enabling automatic retraction of unused screen areas above the model. This process occurs in real-time and is applicable to any model, including 3D scan data.

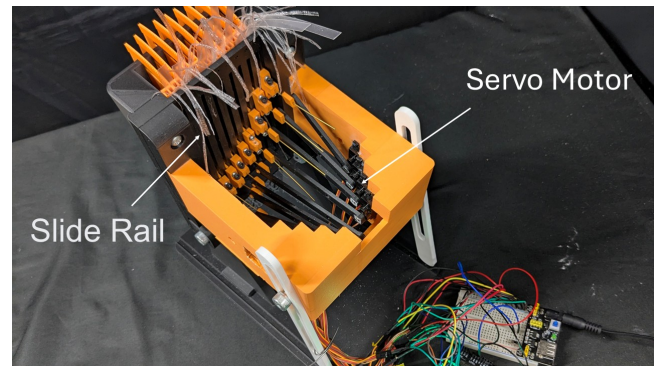


Figure 5: The thread height control mechanism unit with 9 servo motors.

4 Results

Our experiment confirmed stable screen formation with high visibility (Figure 1). The retro-reflective paint provided clear image reflection in dark environments. The height control system demonstrated a real-time response, successfully retracting unnecessary areas based on Z-axis data from Unreal Engine.

Visible vertical segmentation occurred with finer threads but did not significantly hinder image recognition. Mitigating airflow-induced swaying and tangling is expected to improve resolution.

Current challenges include image distortion from thread movement due to unstable airflow and the trade-off between thread density for higher resolution and the increased risk of tangling.

The Thread Display is inherently safe as it uses no liquids or particles and holds promise for haptic integration. Future work will focus on thread directionality, parallax presentation, and creating haptic experiences.

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